

RILEY SEOW

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SUMMARY

Frontend-focused software engineer who works with data-heavy systems to build performant and scalable web applications. Experience working in React/Typescript, D3, Node, and SQL. Adept at integrating diverse data modalities, writing clean, efficient code, and collaborating in cross-functional teams. Concentrated in computational neuroscience via a B.S. in Symbolic Systems from Stanford.

SOFTWARE EXPERIENCE

Software Engineer (UMD Faculty Assistant)

Oct. 2023 - Nov. 2025

Center for Advanced Transportation Technology Laboratory, College Park, MD

- Delivered client- and server-side contributions across three codebases: querying data from servers, databases, and external APIs, and implementing efficient processing pipelines.
- Built high-performance algorithms to process and visualize live data across map layers (e.g. traffic incidents and events, nationwide cameras, dynamic signs, transit, probe speed), handling hundreds of thousands of feature points and geometries refreshed up to every 15 seconds.
- Served as a key contributor to a multi-million-dollar revamp of the RITIS map, a real-time platform used across federal and state governmental agencies.
- Mentored undergraduate and graduate student interns.

Technical Research Intern

Jun. 2020 - Aug. 2020

Stanford Center for Spatial and Textual Analysis, Stanford, CA

- Built a Python web scraper to extract, clean, and categorize character data from Schmoop and LitCharts for a project exploring textual personhood in English classic novels.
- Automated data transformation and export into structured CSV format for downstream use by the word-association modeling team.
- Created scalable, reproducible data pipelines to enable subsequent NLP analysis and modeling.

Research Assistant

Nov. 2019 - Mar. 2020

Stanford Parallel Distributed Processing (PDP) Lab, Stanford, CA

- Developed Javascript code for a web-based speed arithmetic task deployed on Amazon Mechanical Turk to collect behavioral psychology data at scale.
- Collaborated with a PhD researcher to begin evaluating neural network training approaches on the collected dataset, supporting cognitive modeling efforts at the intersection of neuroscience and machine learning.

Research Assistant

Jun. 2019 - Aug. 2019

UCLA Blair Lab, Los Angeles, CA

- Wrote code for an open-source GUI analyzing in-vivo calcium brain imaging data collected using UCLA Miniscope technology.
- Implemented high-performance data visualization algorithms in C# to render real-time neural activity, including max image projections, cell contour overlays, and calcium tracing graphs.

SKILLS

React, Typescript, ES6 JS/HTML/SCSS, D3, Node.js, PostgreSQL, Python, Mapbox, PostGIS, Git, REST APIs, Figma

EDUCATION

Stanford University

B.S. Symbolic Systems | GPA: 4.003 / 4.3